

Last Time

Contrasts

Factorial ANOVA

Partitioning the Variance

Simple Effects

Review of Effect Sizes

Today

Effect Sizes

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Repeated Measures ANOVA

Partitioning the Variance

Post-hoc analysis

Effect Size

A Few Words About Regression

Review: Effect Sizes

η^2 is biased: it overestimates the proportion of “explained” variance.

ω^2 is a correction to η^2 :

$$\omega^2 = \frac{SS_{\text{Between}} - (k - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}}$$

Review: Effect Sizes

η^2 is biased: it overestimates the proportion of “explained” variance.

ω^2 is a correction to η^2 :

$$\begin{aligned}\omega_{\text{Age}}^2 &= \frac{SS_{\text{Age}} - (2 - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}} \\ \omega_{\text{Task}}^2 &= \frac{SS_{\text{Task}} - (5 - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}} \\ \omega_{\text{Age} \times \text{Task}}^2 &= \frac{SS_{\text{Age} \times \text{Task}} - (5 - 1)(2 - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}}\end{aligned}$$

ω^2

```
> omega_squared(eysenck, partial=FALSE)
```

```
# Effect Size for ANOVA (Type I)
```

Parameter	Omega2	95% CI
task	0.55	[0.43, 1.00]
age	0.09	[0.02, 1.00]
task:age	0.06	[0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].

Two-way ANOVA

$$SS_{\text{Total}} = \sum_{k=1}^K \sum_{j=1}^J \sum_{i=1}^{N_{jk}} (X_{ijk} - \bar{\bar{X}})^2$$

$$SS_{\text{Cells}} = \sum_{k=1}^K \sum_{j=1}^J N_{jk} (\bar{X}_{jk} - \bar{\bar{X}})^2$$

$$SS_J = \sum_{j=1}^J N_{j\cdot} (\bar{X}_{j\cdot}^2 - \bar{\bar{X}})^2$$

$$SS_K = \sum_{k=1}^K N_{\cdot k} (\bar{X}_{\cdot k}^2 - \bar{\bar{X}})^2$$

$$SS_{JK} = SS_{\text{Cells}} - SS_J - SS_K$$

$$SS_{\text{Within}} = SS_{\text{Total}} - SS_{\text{Cells}}$$

Three-way ANOVA

See <https://conjugateprior.org/2013/01/formulae-in-r-anova/> for help with formulas.

```
> data <- read.csv("GSS_1980.csv")
> # Convert to factors
> happy <- as.factor(data$HAPPY)
> health <- as.factor(data$HEALTH)
> attitude <- as.factor(data$YOUNGEN)
> age <- data$AGE

> # Fit the ANOVA model
> gss.model <- aov(age~(happy + health + attitude)^3)
> summary(gss.model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
happy	2	3555	1778	9.812	6.12e-05	***
health	3	4746	1582	8.731	1.04e-05	***
attitude	1	5490	5490	30.304	4.89e-08	***
happy:health	6	679	113	0.624	0.7108	
happy:attitude	2	1591	795	4.390	0.0127	*
health:attitude	3	529	176	0.973	0.4050	
happy:health:attitude	5	490	98	0.541	0.7453	
Residuals	850	153997	181			

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Partitioning the Variance: Independent Samples

Recall

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

Model

$$X_{ij} = \mu_j = \mu + \tau_j + \epsilon_{ij}$$

Also

$$E(F) \approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} = \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2}$$

Partitioning the Variance: Independent Two-Way ANOVA

Recall

$$\begin{aligned}SS_{\text{Total}} &= SS_{\text{Cells}} + SS_{\text{Within}} \\&= SS_J + SS_K + SS_{JK} + SS_{\text{Within}}\end{aligned}$$

Partitioning the Variance: Independent Two-Way ANOVA

Recall

$$\begin{aligned}SS_{\text{Total}} &= SS_{\text{Cells}} + SS_{\text{Within}} \\ &= SS_J + SS_K + SS_{JK} + SS_{\text{Within}}\end{aligned}$$

The main effects and interaction are

$$\begin{aligned}\alpha_j &= \mu_{.j} - \mu \\ \beta_k &= \mu_{k.} - \mu \\ \alpha\beta_{kj} &= \mu_{kj} - \mu_{k.} - \mu_{.j} + \mu\end{aligned}$$

Model

$$X_{ijk} = \mu_{jk} = \mu + \alpha_j + \beta_k + \alpha\beta_{jk} + \epsilon_{ijk}$$

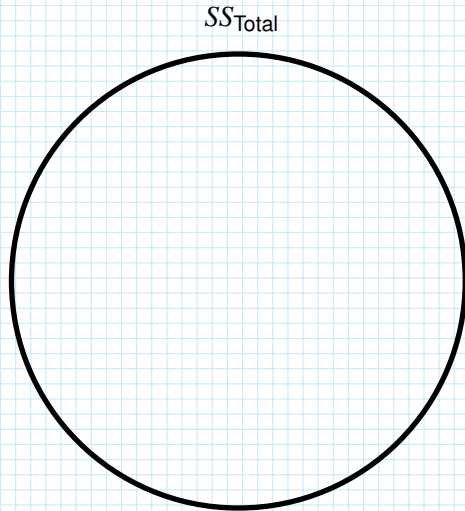
F-ratios for Two-Way ANOVA

$$E(F_J) \approx \frac{E(MS_J)}{E(MS_{\text{Within}})} = \frac{\sigma_e^2 + \frac{\sum_{j=1}^J N_{.j} \alpha_j^2}{J-1}}{\sigma_e^2}$$

$$E(F_K) \approx \frac{E(MS_K)}{E(MS_{\text{Within}})} = \frac{\sigma_e^2 + \frac{\sum_{k=1}^K N_{k\cdot} \beta_k^2}{K-1}}{\sigma_e^2}$$

$$E(F_{JK}) \approx \frac{E(MS_{JK})}{E(MS_{\text{Within}})} = \frac{\sigma_e^2 + \frac{\sum_{j=1}^J \sum_{k=1}^K N_{jk} \alpha_j \beta_{jk}}{(J-1)(K-1)}}{\sigma_e^2}$$

One-way vs. Two-way vs. Repeated measures ANOVA



Dependent groups

- ▶ Same people are in each group *or*
- ▶ Group members are matched on some attribute

Example

Recognition Memory: a list of words is presented one at a time for study. Then a test is presented, where the participant has to say “yes” or “no” to each test item depending on whether it was seen before. Measure correct responses out of 80 words.

Participant	Study Time		
	500 ms	1000 ms	1500 ms
1	43	52	62
2	47	65	75
3	62	71	79
4	55	67	71
5	36	46	57

Each *participant* adds variation because of their different levels of memory ability.

Statistics

Study Time				Person Mean $\bar{P}_i$
Participant	500 ms	1000 ms	1500 ms	
1	43	52	62	52.33
2	47	65	75	62.33
3	62	71	79	70.67
4	55	67	71	64.33
5	36	46	57	46.33
$\bar{X}_j$	48.60	60.20	68.80	$\bar{\bar{X}} = 59.20$

Repeated Measures ANOVA

$$SS_{\text{Total}} = SS_{\text{Between Participants}} + SS_{\text{Within Participants}}$$

and

$$SS_{\text{Within Participants}} = SS_J + SS_{\text{Error}}$$

- ▶ $SS_{\text{Between Participants}}$ is variation due to error and participant factors
- ▶ SS_{Error} is variation due to the way that individual participants respond to each treatment
- ▶ Neither $SS_{\text{Between Participants}}$ nor SS_{Error} can be used to estimate σ_e^2 alone. There is a *participant by treatment interaction* that cannot be separated out.

Partitioning the Variance (Skip)

Define $\bar{P}_i$ to be the mean *across treatments* for Participant i . There are N participants, so N participants in each condition.

$$\begin{aligned}SS_{\text{Total}} &= \sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{\bar{X}})^2 \\&= \sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i + \bar{P}_i - \bar{\bar{X}})^2 \\&= \sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i)^2 + J \sum_{i=1}^N (\bar{P}_i - \bar{\bar{X}})^2 \\&= SS_{\text{Within Participants}} + SS_{\text{Between Participants}}\end{aligned}$$

Partitioning the Variance (Skip)

$$\begin{aligned}SS_{\text{Total}} &= SS_{\text{Within Participants}} + SS_{\text{Between Participants}} \\SS_{\text{Within Participants}} &= \sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i)^2 \\&= \sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i + \bar{X}_j - \bar{X}_j + \bar{\bar{X}} - \bar{\bar{X}})^2 \\&= \sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i - \bar{X}_j + \bar{\bar{X}})^2 + N \sum_{j=1}^J (\bar{X}_j - \bar{\bar{X}})^2 \\&= SS_{\text{Error}} + SS_J\end{aligned}$$

Partial ANOVA Summary Table

Source	df	SS
Between Participants	$N - 1$	$J \sum_{i=1}^N (\bar{P}_i - \bar{\bar{X}})^2$
Within Participants	$N(J - 1)$	$\sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i)^2$
(Treatment) J	$J - 1$	$N \sum_{j=1}^J (\bar{X}_j - \bar{\bar{X}})^2$
Error	$(N - 1)(J - 1)$	$\sum_{j=1}^J \sum_{i=1}^N (X_{ij} - \bar{P}_i - \bar{X}_j + \bar{\bar{X}})^2$

Repeated Measures ANOVA Model

Let π_i be the “effect” of being Participant i . Then

$$X_{ij} = \mu + \pi_i + \tau_j + \pi\tau_{ij} + \epsilon_{ij}$$

- ▶ π_i is another source of error: $\pi_i \sim \mathcal{N}(0, \sigma_\pi)$
- ▶ $\pi\tau_{ij}$ is random: $\pi\tau_{ij} \sim \mathcal{N}\left(0, \frac{J-1}{J} \sigma_{\pi\tau}^2\right)$

Why Am I Doing This to You?

$$E(MS_{\text{Between Participant}}) = \sigma_e^2 + J\sigma_\pi^2$$

$$E(MS_J) = \sigma_e^2 + \sigma_{\pi\tau}^2 + N \frac{\sum_{j=1}^J \tau_j^2}{J-1}$$

$$E(MS_{\text{Error}}) = \sigma_e^2 + \sigma_{\pi\tau}^2$$

How must the F ratio be formulated?

Statistics

Participants	Study Time			$\bar{P}_i$
	500 ms	1000 ms	1500 ms	
1	43	52	62	52.33
2	47	65	75	62.33
3	62	71	79	70.67
4	55	67	71	64.33
5	36	46	57	46.33
$\bar{X}_j$	48.60	60.20	68.80	$\bar{\bar{X}} = 59.20$

In R

```
> recog <- c(
  43, 52, 62,
  47, 65, 75,
  62, 71, 79,
  55, 67, 71,
  36, 46, 57)
> participant <- as.factor(rep(1:5,each=3))
> times <- as.factor(rep(c(500,1000,1500),times=5))
> summary(aov(recog~times+Error(participant/times)))
```

Error: participant

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Residuals	4	1141	285.3		

Error: participant:times

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
times	2	1027.6	513.8	68.81	9.11e-06 ***
Residuals	8	59.7	7.5		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Post-Hoc Analysis: Tukey's HSD

For a one-way ANOVA:

$$HSD = q(\alpha; df_{\text{Within}}, J) \sqrt{\frac{MS_{\text{Within}}}{N}}$$

Replace MS_{Within} with MS_{Error} and df_{Within} with df_{Error} :

$$MS_{\text{Error}} = 7.50$$

$$df_{\text{Error}} = 8$$

$$J = 3$$

$$N = 5$$

$$\alpha = .05$$

$$\rightarrow q(\alpha; df, J) = 4.04$$

$$HSD = q(\alpha; df_{\text{Error}}, J) \sqrt{\frac{MS_{\text{Error}}}{N}} = 4.04 \sqrt{\frac{7.50}{5}} = 4.95$$

$$HSD = 4.95$$

	$\bar{X}_1 = 48.60$	$\bar{X}_2 = 60.20$	$\bar{X}_3 = 68.80$
$\bar{X}_1$	0	11.60	20.20
$\bar{X}_2$		0	8.60
$\bar{X}_3$			0

Note: The R function TukeyHSD() will not work with repeated measures!

Effect Size

η^2 and ω^2 now are problematic.

- ▶ $df_J = J - 1$

- ▶ Partial $\eta^2 = \frac{df_J F}{df_J F + df_{\text{Error}}}$

Effect Size

η^2 and ω^2 now are problematic.

- ▶ $df_J = J - 1$

- ▶ $Partial \eta^2 = \frac{(J - 1)F}{(J - 1)F + df_{Error}}$

Cohen's d still works well.

$$\hat{d} = \left| \frac{\bar{X}_i - \bar{X}_j}{s} \right| = \left| \frac{\bar{X}_i - \bar{X}_j}{\sqrt{MS_{Error}}} \right|$$

A Few Words About Regression

$$\hat{Y}_i = mX_i + b \rightarrow Y_i = mX_i + b + \epsilon_i$$

How does ANOVA fit into this framework?

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$$\hat{Y}_i = mX_i + b \rightarrow Y_i = mX_i + b + \epsilon_i$$

How does ANOVA fit into this framework?

- ▶ Predictor variable X has J levels.
- ▶ The ANOVA model says

$$Y_{ij} = \mu + \tau_j + \epsilon_{ij}.$$

- ▶ Rewrite X_i as a sequence of $J - 1$ *dummy variables* $X_{i1}, X_{i2}, \dots, X_{i,J-1}$ such that

$$X_{ij} = \begin{cases} 1 & \text{if Participant } i \text{ is in level } j \\ 0 & \text{otherwise.} \end{cases}$$

ANOVA is Regression

$$\begin{aligned}Y_{ij} &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{J-1} X_{i,J-1} + \epsilon_{ij} \\ &= \beta_0 + \beta_j + \epsilon_{ij}\end{aligned}$$

- ▶ $H_0 : \mu_1 = \cdots = \mu_J \rightarrow H_0 : \beta_j = 0, j = 1, \dots, J - 1$
- ▶ Level J is the comparison or control condition. If Participant i is in level J , the values of all the dummy variables are 0 and $\beta_0 = \mu_J$.
- ▶ The ANOVA becomes an analysis of the regression weights.